

26CBCA0	FUNDAMENTALS OF COMPUTER SCIENCE	Category	L	T	P	Credits	TE Type
		PCC	2	0	2	3	TCP-P

Preamble

This course introduces computational problem solving and structured programming using the C programming language. It develops algorithmic thinking, logical reasoning, and fundamental programming skills to solve real-world computational problems.

Prerequisite

NIL

Course Outcomes

On the successful completion of the course, students will be able to

No.	Course Outcome	TCE Proficiency Scale	Performance Indicator	CO Weightage	SDG Addressed	SDG Level
CO1	Apply fundamental ANSI C programming concepts and control structures to develop computational solutions for problem-solving tasks	3	1.3.1	15%	4	1
CO2	Demonstrate functions, recursion, parameter passing, and arrays to develop modular data-processing programs	3	1.8.1	20%	8	1
CO3	Solve programming problems using pointer concepts and string operations involving arrays, strings, and functions.	3	1.4.1	20%	8	1

CO4	Use structures, unions, pointers, and file operations to solve complex data organization and processing problems.	3	1.4.1	25%	8	1
CO5	Demonstrate effective technical communication skills through structured documentation, code presentation, and problem-solving activities.	3	9.1.2	5%	NC	-
CO6	Apply teamwork skills to develop, and test, C programming projects collaboratively	3	8.1.1	10%	NC	-
CO7	Evaluate C programming solutions based on professional ethics, software reliability, and sustainable computing practices aligned with relevant SDGs.	3	7.2.1	5%	NC	-

Assessment Pattern

CO	Theory		Practical		Terminal Exam % of Marks Practical
	CAT1	CAT2	OCR/Continuous Assessment	Model Test CO-wise	
	% of Marks	% of Marks	% of Marks	% of Marks	
CO1	40		15	15	
CO2	60		20	20	
CO3		50	20	20	

CO4		50	25	25	100
CO5			5	5	
CO6			10	10	
CO7			5	5	
Total	100	100	100	100	100

Assessment Pattern: Practical Component

Type of Assessment	Component	Marks	Remarks
Continuous Assessment	Pre-lab, In-lab, Post-lab	75	Conducted throughout the semester
Model Test	CO-wise (Cognitive)	25	Rubrics covering all Domains of learning
Total Internal Marks		100	
End Semester Examination	CO-wise (Cognitive)	100	Rubrics covering all Domains of learning

Syllabus

Problem Solving Concepts: Fundamentals of ANSI C Programming – Structure of a C program – Compilation and execution process – variables, constants, data types – scope and lifetime of variables - Operators, Expressions and Type conversion – Precedence and Order of Execution - break, continue, exit, goto labels – Basic Syntax and Logical Errors in compilation, source, object and executable code

Control Structures: Statements and blocks – if, if-else, else-if ladder and switch – while, do-while and for loops – Nested loops

Functions: Function prototype, function definition, function call, Built-in functions – Recursion – Parameter passing: Pass by value, Pass by reference - Preprocessor directives and macros – Standard library functions

Arrays and Strings: Declaration, Initialization – One-dimensional array – Two-dimensional arrays – Passing arrays to functions - maximum and minimum value – Selection Sort – Linear and Binary search techniques - Character Array - String operations (length, concat, compare, copy)

Pointers: Pointer arithmetic – Arrays and pointers – Character pointers and strings – Pointer arrays and pointer to pointer – Pointers to functions

Structures and Union: Structures and functions – Arrays and pointers to structures – Self-referential structures – Array of Structures - typedef and bit-fields

Files: Introduction to files - Types of file processing: Sequential access, Random access - Command line arguments– Error handling

Text Book

1. Yashavant Kanetkar, "Let Us C", BPB Publications, 21st edition, 2025.

Reference Books & web resources

1. Jens Gustedt, Modern C, 3rd Edition, Manning Publications, 2025.

- Herbert Schildt, "C: The Complete Reference", Fourth Edition, McGraw Hill, 2017
- Brian W. Kernighan and Dennis M. Ritchie, The C Programming Language, 2nd Edition, Prentice Hall / Pearson Education, 1988.
- NPTEL – " Problem Solving through Programming in C" -
https://onlinecourses.nptel.ac.in/e-learning/preview/noc26_cs53

SDG Alignment

Activities aligned with SDGs addressed:

- SDG 4 – C Programs that enables the students to convert the mathematical concepts into digital codes
- SDG 8 – Programs using arrays, pointers and string will help to analyze the utilization of systems memory and improvise it effectively.

Course Contents and Lecture Schedule

Module No	Topics	No. of Periods	Course Outcomes
1	Problem Solving Concepts & Control Structures		
1.1	Fundamentals of ANSI C Programming – Structure of a C program – Compilation and execution process - variables, constants, data types	1	CO1
1.2	Scope and lifetime of variables		
1.3	Operators, Expressions and Type conversion – Precedence and Order of Execution - break, continue, exit, goto labels	1	CO1
1.4	Basic Syntax and Logical Errors in compilation, source, object and executable code		
1.5	Statements and blocks – if, if-else, else-if ladder and switch	1	CO1
1.6	while, do-while and for loops – Nested loops	1	CO1, CO7
2	Functions and Arrays:		
2.1	Function prototype, function definition, function call, Built-in functions – Recursion	1	CO2
2.2	Parameter passing: Pass by value, Pass by reference -	1	CO2
2.3	Preprocessor directives and macros – Standard library functions	1	CO2
2.4	Declaration, Initialization – One-dimensional array –Two-dimensional arrays - Passing arrays to functions	1	CO2, CO6
2.5	Maximum and minimum value, Selection Sort – Linear and Binary search techniques	1	CO2, CO5
3	Strings and Pointers		
3.1	Character Array - String operations (length, concat, compare, copy)	1	CO2
3.2	Pointer arithmetic – Arrays and pointers	1	CO3
3.3	Character pointers and strings	1	CO3
3.4	Pointer arrays and pointer to pointer	1	CO3, CO6
3.5	Pointers to functions	1	CO3
4	Structures, Union and Files:		
4.1	Structures and functions	1	CO4, CO5

4.2	Arrays and pointers to structures – Self-referential structures	2	CO4
4.3	Array of Structures - typedef and bit-fields	1	CO4
4.4	Introduction to files - Types of file processing: Sequential access, Random access -	1	CO4, CO6
4.5	Command line arguments– Error handling	1	CO4, CO7
Total Periods		24	

List of Experiments with CO Mapping

Sl. No.	Name of the Experiment	Level of Experiment	Objective(s)	COs Addressed
1	Develop a C program to perform arithmetic and relational operations on different data types and analyze the effect of type conversion and operator precedence	1	3- Experimentation	CO1
2	Construct a menu-driven application using conditional statements to perform different operations based on user choice.	1	3- Experimentation	CO1, CO5
3	Develop a C program using loops to generate number patterns and perform mathematical computations such as factorial and Fibonacci series.	1	2- Models	CO1
4	Design a C program to solve mathematical problems using both user-defined functions and recursive techniques.	2	3- Experimentation	CO2
5	Develop a C program to compare pass by value and pass by reference methods for performing data manipulation operations.	2	3- Experimentation	CO2, CO5
6	Construct a C program to perform matrix operations and determine the maximum and minimum elements in arrays.	2	2- Models	CO2
7	Develop a C program to sort a dataset and search for a specific element using suitable searching techniques.	2	2- Models	CO2
8	Design a C program to perform string operations such as concatenation, comparison,	2	3- Experimentation	CO3, CO7

	and copying without using built-in library functions.			
9	Develop a C application using pointers and function pointers to process arrays and perform dynamic function execution.	3	3- Experimentation	CO3, CO6
10	Construct a file-based C application using structures to store, retrieve, and manage records efficiently.	3	3- Experimentation	CO4, CO6, CO7
11	MINI PROJECT*	3	9 - Design	CO1 – CO7

***Assessment:** The Mini Project carries a 10% weightage in the Test marks. Starting from Exercise 9, students will do the exercises based on their chosen Mini Project. Students must work in teams of three members for the Mini Project. It will be evaluated based on all CO's.

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